Measuring Emotion:
Self-Reports vs. Physiological Indicators

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Prepared for the 2015 Annual Meeting of the Midwest Political Science Association

Abstract

While emotion has gained popularity in political science research, its measurement is still an open question. To date, most work on emotion and behavior relies on survey respondents to accurately and honestly report their emotional responses to relevant stimuli. While this strategy is relatively inexpensive and efficient, its accuracy has been called into question. More specifically, it has been suggested that people (1) have a difficult time pinpointing specific reasons for their attitudes, (2) hesitate to give honest answers when such answers are socially undesirable, and (3) rationalize their answers to such questions for various reasons. An alternative measurement strategy involves recording respondents' physiological responses to stimuli, which indicate activation of the autonomic nervous system. Unlike self-reports, physiological measures are not subject to social desirability bias, and they can capture aspects of emotional response that are beyond respondents' conscious control. Despite a wealth of research from both political science and psychology on the measurement characteristics of emotion self-reports and a large body of psychological research on physiological measures, there is a lack of understanding on the degree to which the two types of measures "converge." Our purpose here is two-fold. First, we examine the extent to which physiological reactions to stimuli influence self-reports. Second, we look for conditions under which one measure outperforms the other in terms of explanation/prediction. In short, we find that self-reports are better predictors of political attitudes, but also, self-reports may provide biased estimates of emotions' effects on attitudes in respondents that score high on emotion regulation scales.

In recent years, the study of emotion has gained popularity in political science research. Scholarship shows that emotions affect a wide range of political behaviors, including attitude formation (Aarøe 2011; Gross 2008; Gross and D'Ambrosio 2004), information processing (Marcus et al. 2000), policy preferences (Westen 2010), ideology (Oxley et al. 2008), and participation (Gruszczynski et al. 2013; Valentino et al. 2011). To date, most political scientists have relied on respondents to accurately and honestly report emotional responses to given stimuli. And, while this measurement strategy is inexpensive and efficient (McDermott 2007), its accuracy has been called into question. More specifically, Krosnick (1988) suggests that most people have a difficult time pinpointing specific reasons for attitudes and actions. Berinsky (2004) specifies conditions under which respondents might hesitate to give honest answers regarding preferences or attitudes –when such answers are socially undesirable. Finally, McDermott (2007) notes that self-reported emotional responses may be rationalizations for their evaluation of the stimuli under consideration. An alternative strategy –using physiological responses as indicators of activation of the autonomic nervous system (ANS)– while popular among psychologists (e.g. Troy et al. 2013), is only starting to make inroads in political science (e.g. Oxley et al. 2008). Without doubt, much has been done to better understand each form of measurement, their strengths, and their weaknesses, but little has been done to weave the two measurement techniques together.

Our purpose, stated briefly, is two-fold. First, we intend to shed light on how respondents' physiological reactions to policy stimuli influence their emotion self-reports. Second, we look for conditions under which one measurement technique outperforms the other in terms of explanation/prediction. Using data gathered on 106 undergraduates, we show that emotion self-reports explain a great deal of variance on attitudes relating to federalism and immigration policy. We also show that physiological response is mediated through emotion self-reports, and that it does have some indirect influence on expressed

attitudes. Last, we discuss the ways in which individual differences in social desirability and emotion regulation affect the measures.

Past Research

Early social psychological work on measuring emotion started with description and worked toward finding the "structure of emotion." By using multiple measures of many different emotions, researchers sought to accurately name emotions and locate their factor dimensions (Russell and Carroll 1999a,b; Watson and Tellegen 1999). The general consensus, at least in social psychology, is that two orthogonal dimensions are needed to adequately describe response structure (Fabrigar et al. 1997; Remington et al. 2000). There is disagreement, however, on the interpretations of the two-factor solutions. On one hand, the valence approach identifies one dimension as pleasurable/unpleasurable, and the other as arousal (Clore and Ortony 2000). On the other, scholars identify the two primary axes as positive affect and negative affect (Cacioppo et al. 1997, 1999). With respect to measurement, the implications of the two theories are straightforward. Insofar as the valence model ignores the "arousal" dimension, one only needs one item per stimulus to measure emotional response (e.g. a feeling thermometer). The "positive/negative" interpretation, however, requires two items per stimulus –one to assess positive emotions and another to assess negative (Marcus et al. 2006).

Despite the prolonged debate in psychology literature, a good deal of political science research on emotion and behavior deviates from the above-mentioned recommendations. Marcus et al. (2006) note that the emotions of politics may be more distinct and complex than those they encounter in daily life (p. 34). This so, the authors recommend that political scientists allow for more than two dimensions when necessary (see, for example Valentino et al. 2011). Further, because negative and positive emotions (often referred to as anxiety and enthusiasm) arise from two different sources in the brain (Marcus et al.

2000, chaps. 3 & 4), because they can be experienced simultaneously, and because they can be directed at the same target (e.g. one can simultaneously experience positive and negative emotions toward one's children), it is often the case that researchers allow these dimensions to be correlated (Marcus et al. 2006, 2000).

Despite the progress on survey-based measures of emotional response, there is a large body of research that suggests that emotion self-reports may not provide researchers with valid measurements of relevant psychological processes. Critiquing survey research in general, scholars note that many survey respondents may not be able to pinpoint specific reasons for actions and attitudes (Krosnick 1988), that social desirability may bias response (Berinsky 2004), and that responses may be rationalizations of feelings experienced at a subconscious level (Lodge and Taber 2013). Measures of physiological response, which indicate activation of the autonomic nervous system (e.g. fight or flight), have long been used in psychology to gauge affect, but are only starting to gain popularity in political science (e.g. Gruszczynski et al. 2013; Oxley et al. 2008; Renshon et al. 2014; Soroka and McAdams 2015). As these measures are biological in nature, they are not subject to social desirability or any other associated concern (Mauss and Robinson 2009). Furthermore, they are able to capture what may be beyond a respondent's conscious control and understanding (Larsen et al. 2008).

To what extent do emotion self-reports and physiological responses measure the same concept? It is commonly argued that emotional response involves coordinated changes across a wide variety of systems (Dolan 2002; Levenson 1994; Tomkins 1962). That is to say, emotional response ties experience, behavior, and physiology together. Empirical work on the topic, however, fails to reach such a conclusion. Mauss et al. (2004), using anxiety inductions on high- and low-trait social anxiety participants, assessed the degree to which the anxiety influenced experiential, behavioral, and physiological systems. While the anxiety induction did cause differences in experiential and behavioral measures, there

was no discernable differences on physiological measures. In a similar study, Mauss et al. (2005) find strong associations between experiential and behavior responses, but only moderate associations with physiological response. These findings imply that there may be important differences between the factors that drive responses to emotion self-reports and those that drive physiological reactions.

What explains these differences? In a very broad way, survey measures and physiological responses indicate emotional reactions, but it has been noted that the self-report scales often used in political science research come closer to Damasio's (1994) understanding of *feeling* (McDermott 2007, p. 377), where physiological measures get at *affect*. McDermott also notes that physiological reactions occur prior to self-reports (also see Zajonc 1980, 1984), which could explain inconsistencies in the two types of measures. Finally, while it is possible that physiological reactions inform self-reports, it is also possible that other factors (e.g. partisanship) will influence self-reports as well. That is, self-reported emotions may be rationalizations that serve to preserve other parts of one's identity (Lodge and Taber 2013).

Research Questions and Theoretical Expectations

There are three questions central to this paper. First, to what extent do physiological reactions to political stimuli influence emotion self-reports? Second, to what extent do both measures of emotion explain variance on political attitudes? Third, what individual-level factors might lead to biased emotion self-reports? Here, do physiological measures provide a viable alternative to self-reports?

Our theoretical expectations are guided by past research that suggests first that emotional response involves a wide variety of systems (e.g. Tomkins 1962), but also that stronger emotions produce tighter coupling between systems (e.g. Mauss et al. 2005). With this in mind, we expect physiological response to have a greater impact on self-

reports when the issue in question produces strong emotional reactions. In the same vein, we expect both measures to explain variance on political attitudes when the issue in question produces a strong emotional reaction. Further, we expect physiological reactions to effect reported attitudes directly and indirectly, where positive and negative emotions self-reports act as mediating variables (see figure 1). With respect to our third question, we test the degree to which tendency to engage in socially desirable responding[1] and the degree to which respondents are able to regulate their emotions[2] affects estimates of emotions as they influence attitudes.

[Figure 1 about here]

Data, Measures, and Methods

To answer the questions posed above, we recruited students from introductory level government and psychology classes at Franklin & Marshall College during the spring 2014 semester. Students first filled out a preliminary questionnaire where they answered questions on partisanship, ideology, social desirability, and emotion regulation. The main survey was taken in a lab where we were able to monitor students' skin conductance while they answered questions on federalism, immigration, and various other issues.

Lab Procedure

Upon entering the lab, students sat at a computer and trained research assistants attached sensors to measure skin conductance and heart rate. Skin conductance sensors

[1]Berinsky (2004, chap. 1), for example, argues that survey respondents, rather than provide an honest answer that may be socially undesirable, may opt to provide a polite answer that "saves face." We think that some people may feel the same way when filling out emotion self-reports –that to feel certain emotions on certain sensitive issues is socially undesirable. Therefore, people provide socially desirable answers, thus biasing the measure.

[2]Emotion regulation refers to the set of processes whereby people seek to redirect the spontaneous flow of their emotions, usually to change core affect (Koole 2009). The ability/inability to regulate emotion is often linked with stress, depression, and anxiety (Aldao et al. 2010; Troy et al. 2010). In the context of this paper, we expect the ability to regulate one's emotions to bias emotion self-reports (e.g. those that score high on emotion regulation will systematically mis-report their emotional states).

were placed on the thenar and hypothenar surfaces of the palm of the non-dominant hand. Three ECG sensors were placed on the lower abdomen and collarbone in order to obtain a continuous measure of heart rate. Students were instructed to keep their non-dominant hand as still as possible for the entire session. They were instructed to use their dominant hand to make their responses to questions using the mouse. Once physiological sensors were attached, the students were then asked to click a radio button when they were ready to begin the survey. Next, students were asked to sit quietly for one minute, which was necessary in order to collect baseline data on skin conductance and heart rate. After the one-minute baseline, students were exposed to two policy statements: one on federalism and one on immigration policy (the order in which they were seen was randomized, see appendix for complete wording). Students were asked to read and think about the statement for one minute (again to collect data on skin conductance and heart rate) before they were allowed to answer questions on whether they agreed or disagreed with the statement. Subjects were then asked about the emotions they felt when reading the policy statements.

Dependent Variables

The two main dependent variables for the analyses in this paper measure the extent to which subjects agreed/disagreed with the two policy statements. Each variable is measured on a 7-point scale where 1 indicates "strongly disagree" and 7 indicates "strongly agree."

Emotion Self-Reports

After indicating agreement/disagreement with the policy statement, subjects were asked about the extent to which the policy statement made them happy, enthusiastic, proud, sad, angry, fearful, and disgusted (all items were coded on a 1-5 scale where 1

indicates they did not feel the emotion and 5 indicates they strongly felt the emotion).

Following the recommendations of Marcus and colleagues, we ran a factor analysis (promax rotation) on the data, expecting to find two factors corresponding to negative and positive emotions. For both issues, the model fit the data very well. The confirmatory factor analysis on emotion self-reports related to the federalism issue suggests that the two-factor model is an adequate representation of the data (CFI = .961, TLI = .938, RMSEA = .074), and when allowing "happy" and "sad" to be correlated, the fit improves (CFI = .967, TLI = .943, RMSEA = .071).[3] The two-factor model also fit the emotion self-reports related to the immigration issue (CFI = .966, TLI = .946, RMSEA = .093).[4] It is also worth noting that the promax rotation is necessary: on the federalism issue, the two factors are correlated at -0.37; on the immigration issue, the two factors are correlated at -0.41.[5] For the purposes of analysis, the factor scores were converted to z-scores.

Skin Conductance

Physiological reactions (skin conductance and heart rate) were measured with Mindwares Bionex 8-slot chassis. All data were recorded using Biolab Acquisition Software (Mindware Technologies LTD; Gahanna, OH). Following data collection, Mindwares EDA 3.0.21 program was used for data reduction, artifact control, and the computation of average skin conductance level (SCL) for each participant for each segment of the experiment. Change in skin conductance level (ΔSCL) was calculated by subtracting subjects' baseline level from the level measured during exposure to the stimulus. All measures were converted to z-scores.

[3]We tested a one-factor model and three-factor model as well. The two-factor model clearly fit better than the alternatives.

[4]Again, we tested a one- and three-factor model as well. Here too, the fit of the two-factor model was superior to the alternatives.

[5]For both issues, we examined unrotated solutions, varimax rotations, and promax rotations. And, for both issues, the promax rotation produced "cleaner" (i.e. larger in magnitude and closer to one) factor loadings.

After removing subjects from the data set for various reasons,[6] we were left with 77 complete observations on the federalism issue and 73 complete observations on the immigration issue (53 and 55 percent female, respectively).

Other Measures

In addition to the variables mentioned above, we also collected data on partisan identity (three dummy variables corresponding to Republican identification, Democratic identification, and independent), race (coded such that 1 = nonwhite and 0 = white), and sex (coded such that 1 = female and 0 = male). Tendency to provide socially desirable responses is measured with a 13-item battery from the Marlowe-Crowne Social Desirability Scale ($\alpha = .66$). We included three measures of emotion control: general emotion control is measured with a 2-item scale ($r = .52$), emotion reappraisal is measured with a 7-item scale ($\alpha = .85$), and emotion suppression is measured with a 5-item scale ($\alpha = .64$).

Methods

In looking at our first research question (to what extent do physiological reactions influence emotion self-reports?), we using a seemingly unrelated regression (SUR) model to regress negative/positive emotion self-reports on skin conductance level and a host of control variables. The SUR specification increases statistical efficiency, and because the models are estimated simultaneously, coefficients can be compared across models (Zellner 1962).

Second, we examine the degree to which each type of emotion measure explains variance on political attitudes with OLS models and causal mediation analysis. From Ren-

[6] If subjects removed the electrodes from their hand for any reason during the session (e.g. uncomfortable, fell off, etc.), they were removed from the data set. Also, following (Renshon et al. 2014), we deleted subjects that experienced extraordinarily wide swings in skin conductance levels from pre-test to post-test (i.e. those for whom ΔSCL is $+3$ or -3 standard deviations away from the mean).

shon et al. (2014, p. 10), causal mediation analysis allows one to estimate the average causal mediation effect (ACME), which can be interpreted as the expected difference in the outcome variable when the mediator (in this case, positive and negative emotion self-reports) takes the value it would have under the under the "treatment" condition as opposed to the "control" condition.[7] Though we never observe the outcome in both states, we are able to estimate this effect as long as the sequential ignorability assumption is satisfied.[8]

Finally, we use OLS models with interaction terms to estimate the effects of social desirability and the various types of emotion regulation as they effect self-reports. We first compare the fit statistics of models with and without interaction terms to determine whether social desirability and emotion regulation affect measurement at all. We then use the coefficients on the interaction terms to determine the extent to which tendency to provide social desirable responses and emotion regulation affect emotion self-reports.

Results

To what extent do physiological reactions influence emotion self-reports? The results of the SUR models are shown in tables 1 and 2. Looking first at the results of the model of emotion self-reports on the federalism issue, the only variable with any significant explanatory power is sex: females, on average, report feeling more positive emotion, and less negative emotion, than males. The results of the model of emotion self-reports on immigration suggest that physiological reactions may affect self-reports, but the results only

[7] Obviously, this is written for experimental data. For our purposes, the "control condition" is $\Delta SCL = 0$. Our "treatment," which is the change in SCL from the resting period through exposure to the stimulus, is continuous.

[8] Sequential ignorability first assumes that the treatment (i.e. ΔSCL) is statistically independent (that is, ignorable) of potential outcomes and mediators. Because physiological reactions occur at a subconscious level, and because these reactions occur before one makes decisions on a policy or before one is conscious of the emotions they are feeling, we think this assumption holds. Second, sequential ignorability assumes that the mediators (i.e. positive and negative self-reports) are ignorable as well. Because this is a much stronger assumption, we run sensitivity analysis, as suggested by Imai et al. (2011).

holds for negative emotions. An increase in ΔSCL corresponds, counter to expectations, to a decrease in reported negative emotion. One control variable, the dummy flagging whether or not one identifies as Republican, registers a significant coefficient, suggesting that Republicans, on average, report more positive emotion to the policy statement.

[Tables 1 and 2 about here]

Do physiological reactions to stimuli affect reported attitudes? According to the OLS models (table 3), physiological reactions do not directly affect reported attitudes. The mediation analyses (see figure 2) on attitudes regarding federalism suggest the same – that physiological reactions have neither a direct or indirect affect on reported attitudes. And on the immigration stimulus, while there is no indirect effect of physiology through positive emotion self-reports, there is some evidence of an indirect effect through negative self-reports ($ACME = 0.29$, $p = .06$).[9] The implication here is that, though small, physiological reactions do influence reported attitudes, albeit indirectly.

[Table 3 and figure 2 about here]

How might individual-level characteristics bias emotion self-reports? Recall, there is reason to believe that propensity to give socially desirable responses may affect how one self-reports emotion (e.g. Berinsky 2004), and there is reason to believe that various forms of emotion regulation may influence emotion self-reports (e.g. Troy et al. 2013). The results of our analyses are displayed in tables 4 and 5. Looking first at table 4, only one form of response regulation seems to exert a systematic effect on emotion self-reports as they influence attitudes: general emotion regulation. More specifically, respondents that score high on the general emotion regulation scale tend to be less affected by negative emotion (that is, self-reported negative emotion seems to have a smaller effect on attitudes

[9]The sensitivity analysis concerning the correlation ρ between the residuals of the mediator and outcome regressions suggest that our results are fairly robust to unobserved pre-treatment confounders that might affect both the mediator and the outcome (Tingley et al. 2014, p. 14).

regarding the federalism measure), which suggests that the coefficient on negative self-reports in the baseline model *overestimates* the effects of reported negative emotion for some and *underestimates* the effects for others. Also of note, the fit of this model is significantly better than the fit of the baseline model ($R_1^2 = .395$, $R_2^2 = .491$, $p < .01$). On responses to the federalism issue, no other form of regulation produces significant effects.

Turning attention to table 5, it is apparent that the effects of general emotion regulation are not consistent across issues. And, while no set of interaction terms significantly improves the fit of the models, there is one interaction that reaches significance: emotion reappraisal as it effects negative self-reports. Interestingly, the coefficient suggests that those that score high on appraisal are *more* affected by negative emotion with respect to immigration attitudes. That is, reappraisal seems to exacerbate the effects of negative emotion, which runs counter to most literature on the topic. Again, self-reported data seems to *underestimate* the effects of negative emotion as it concerns immigration attitudes.

[Tables 4 and 5 about here]

Conclusion and Discussion

To what extent does emotion, as measured with physiological reactions, converge with emotion self-reports? Our data suggest that the two forms of measurement are related, but also that the relationship is minimal. Physiological reactions to the federalism stimulus do not seem to affect self-reports; they do, however, seem to have a small effect (in the expected direction) on self-reports of negative emotion pertaining to the immigration stimulus. Further, our mediation analyses suggest that this physiological response has a significant indirect effect on attitudes through emotion self-reports. Overall, we do feel comfortable concluding that emotion self-reports and physiological reactions converge to

some extent, but we also realize that there is more to be done.

Our study includes only two issues. While we believe this is a good start, we also believe that future research should include a wider array of policies. Further, we believe that it would be beneficial to include very "strong" stimuli (i.e. stimuli that produce a strong emotional reaction) to test the conclusions of Mauss et al. (2005) that measures converge to a greater extent when emotional responses are particularly strong.[10]

Do the different measurement techniques have roughly equal levels of predictive validity when it comes to political attitudes? According to our data, the answer is a fairly clear "no." While physiological reactions do explain some variance (and most of that is through indirect effects), self-reports carry the vast majority of the weight. The predictive success of the models reported in table 3 hinge on self-reports. When those two variables are excluded from the models, the R^2 statistics drop to .058 (from .394) and .200 (from .668) for federalism and immigration, respectively.

The discrepancy between the predictive power of self-reports and physiological response suggests, very simply, that the political world is incredibly complex (as in Marcus et al. 2006) and that physiological response, which is an indicator of only the most basic form of emotion, cannot pick up on all of the nuances that come together during attitude formation. Second, the discrepancy opens the possibility that emotion self-reports are rationalizations of previously stated attitudes. Put simply, when two variables account for 85 and 83.5 percent of the explanatory power of the models on federalism and immigration, respectively, one must consider the possibility that all three variables (i.e. policy agreement, positive self-reported emotion, and negative self-reported emotion) are causally connected in a very complex way.

On the whole, our data suggest that emotion self-reports are valid indicators, but

[10]Though both of our stimuli produced physiological reactions that differed from the baseline –the mean ΔSCL for the federalism stimulus was 0.30 μS ($p = .02$); the mean ΔSCL for the immigration stimulus was 0.22 μS ($p = .14$)– it is not the case that the immigration stimulus produced a significantly greater reaction, as was expected.

our regression models (tables 4 and 5 do suggest that there are individual-level factors that may add systematic error to these measures. We tested the extent to which social desirability, general emotion regulation, emotion reappraisal, and emotion suppression affected the estimates of negative and positive self-reports on attitudes, and while our results were not consistent across issues, they do suggest that there is more to be done here. Specifically, our data suggest that general emotion regulation affects the degree to which negative emotion influences policy agreement on federalism, and that emotion reappraisal exacerbates the effects of negative emotion on policy agreement on immigration. These results are certainly helpful, but they raise several questions: On what types of issues are people most likely to regulate their emotions? Under what conditions does reappraisal exacerbate the effects of self-reports? Under what conditions does reappraisal mitigate effects? Are there any measures of emotion that have significant predictive power and are not subject to emotion regulation or social desirability? Future research might consider some of the questions we raise, and answer to these questions might give us a greater understanding of emotion, its measurement, and how emotions affect political behavior.

To summarize, emotion has gained popularity in research on political behavior and psychology. While most research in political science, to this point, has used self-reports to measure emotion (e.g. Valentino et al. 2011), it is argued that such measures may be susceptible to social desirability bias (Berinsky 2004), that they may be inaccurate because people are often unable to pinpoint reasons for their behavior (Krosnick 1988), and that they may be rationalizations (Lodge and Taber 2013). Physiological response, an alternative measure that is now gaining popularity in political science (e.g. Oxley et al. 2008), is not susceptible to social desirability or related concerns (Mauss and Robinson 2009) and can capture what may be beyond respondents' conscious control (Larsen et al. 2008). Though literatures have been built up around each individual measurement technique, little has been done on how the measures converge (Mauss et al. 2004, 2005). We

add to this literature by showing that physiological reactions influence self-reports, and that physiological reactions indirectly influence attitudes through emotion self-reports. In spite of our results regarding physiology influencing self-reports and attitudes, we argue that physiological reactions are not a viable alternative to replace self-reports, at least in the context of traditional public opinion research.

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Figure 1: Hypothesized Mediation Effects

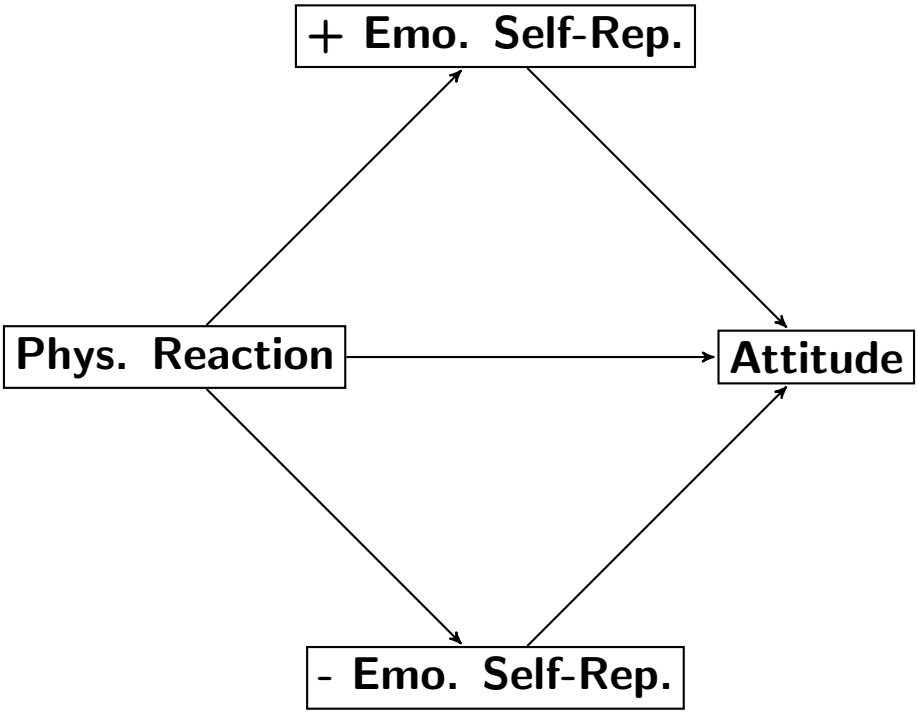


Table 1: SUR Model: Federalism

	Positive Emo.		*Negative Emo.*	
	Coef.	S.E.	Coef.	S.E.
Constant	-0.75	(.462)	0.42	(.454)
SCL	-0.08	(.154)	-0.04	(.115)
Female	0.49**	(.238)	-0.48**	(.234)
Nonwhite	0.02	(.273)	0.25	(.269)
Democrat	-0.01	(.261)	0.13	(.257)
Republican	-0.13	(.345)	0.39	(.339)
R^2	.080		.112	
$r(e_{Pos}, e_{Neg})$	-0.32			
n	77			

Dependent variables are factor scores corresponding to self-reported positive and negative emotion

* $p < .1$; ** $p < .05$; *** $p < .01$

Table 2: SUR Model: Immigration

	Positive Emo.		*Negative Emo.*	
	Coef.	S.E.	Coef.	S.E.
Constant	-0.84*	(.447)	0.35	(.473)
SCL	-0.10	(.119)	-0.23*	(.126)
Female	0.29	(.227)	-0.24	(.240)
Nonwhite	0.33	(.267)	-0.04	(.282)
Democrat	-0.09	(.254)	0.33	(.269)
Republican	0.90***	(.322)	-0.49*	(.340)
R^2	.225		.135	
$r(e_{Pos}, e_{Neg})$	-0.43			
n	73			

Dependent variables are factor scores corresponding to self-reported positive and negative emotion

* $p < .1$; ** $p < .05$; *** $p < .01$

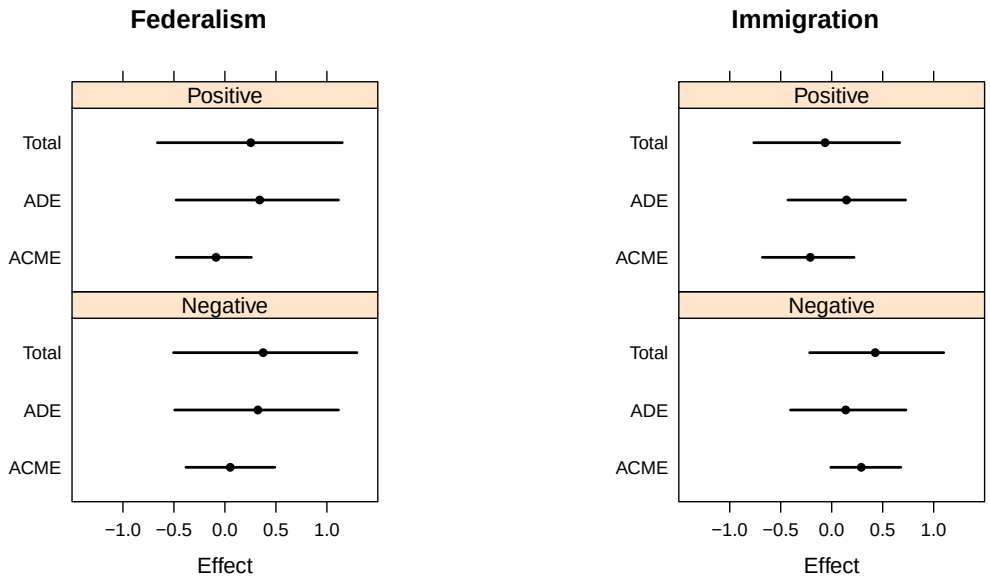
Table 3: OLS Models: Explaining Attitudes on Federalism and Immigration

	Federalism		*Immigration*	
	Coef.	S.E.	Coef.	S.E.
Constant	5.00***	(.557)	2.95***	(.503)
Pos. Self-Report	0.48***	(.148)	0.89***	(.148)
Neg. Self-Report	-0.46***	(.151)	-0.58***	(.140)
SCL	0.14	(.183)	0.06	(.137)
Democrat	-0.24	(.310)	-0.19	(.282)
Republican	-0.00	(.412)	-0.02	(.373)
Female	0.08	(.294)	0.21	(.253)
Nonwhite	0.04	(.326)	0.33	(.296)
R^2	.394		.668	
n	77		73	

Dependent variables are factor scores corresponding to agreement with policy (1 = strongly disagree, 7 = strongly agree)

* $p < .1$; ** $p < .05$; *** $p < .01$

Figure 2: Direct, Indirect, and Total Effects of Physiological Reactions



"Total" refers to total effect, "ADE" refers to average direct effect, and "ACME" refers to average causal mediation effect (i.e. indirect effect). Error bars are 95 percent confidence intervals. Mediation analysis was done with `mediation` package in **R** (Tingley et al. 2014).

Table 4: Effects of Emotion Regulation and Social Desirability on Measurement (Federalism)

	Baseline		*General*		*Reappraisal*		*Suppression*		*Soc. Desirability*	
	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.
Constant	4.95***	(.138)	4.93***	(.129)	4.94***	(.141)	4.95***	(.144)	4.95***	(.139)
Regulation	–		-0.16	(.131)	0.04	(.163)	0.03	(.149)	-0.03	(.141)
Pos. Self-Report	0.48***	(.151)	0.39***	(.147)	0.48***	(.155)	0.48***	(.157)	0.48***	(.153)
× Reg.	–		0.19	(.171)	-0.12	(.185)	0.05	(.154)	0.20	(.180)
Neg. Self-Report	-0.64***	(.151)	-0.74***	(.148)	-0.64***	(.155)	-0.63***	(.164)	-0.60***	(.155)
× Reg.	–		0.49***	(.152)	0.07	(.179)	0.02	(.185)	0.23	(.187)
R^2	.395		.491		.404		.396		.411	
F	–		4.19***		0.36		0.04		0.59	
n	73		73		73		73		73	

From left to right, the type of regulation that is captured by the "regulation" variable, is general emotion regulation (i.e. "general"), emotion reappraisal (i.e. "reappraisal"), emotion suppression (i.e. "suppression"), and social desirability (i.e. "soc. desirability").

The F-statistic was obtained by comparing the fit of baseline model against the fit of the alternative model.

* $p < .1$; ** $p < .05$; *** $p < .01$

Table 5: Effects of Emotion Regulation and Social Desirability on Measurement (Immigration)

	Baseline		*General*		*Reappraisal*		*Suppression*		*Soc. Desirability*	
	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.
Constant	3.52***	(.114)	3.52***	(.114)	3.51***	(.114)	3.53***	(.115)	3.51***	(.116)
Regulation	–		0.04	(.118)	-0.08	(.124)	-0.13	(.125)	-0.11	(.118)
Pos. Self-Report	0.80***	(.127)	0.82***	(.129)	0.87***	(.139)	0.80***	(.131)	0.75***	(.131)
× Reg.	–		0.17	(.127)	-0.00	(.133)	0.09	(.122)	-0.17	(.120)
Neg. Self-Report	-0.79***	(.127)	-0.81***	(.128)	-0.77***	(.138)	-0.78***	(.132)	-0.81***	(.129)
× Reg.	–		-0.07	(.141)	-0.23*	(.139)	-0.02	(.132)	-0.03	(.122)
R^2	.672		.689		.691		.682		.686	
F	–		1.17		1.28		0.64		0.96	
n	69		69		69		69		69	

From left to right, the type of regulation that is captured by the "regulation" variable, is general emotion regulation (i.e. "general"), emotion reappraisal (i.e. "reappraisal"), emotion suppression (i.e. "suppression"), and social desirability (i.e. "soc. desirability").

The F-statistic was obtained by comparing the fit of baseline model against the fit of the alternative model.

* $p < .1$; ** $p < .05$; *** $p < .01$

Appendix A: Question Wordings

Federalism Stimulus

A group of Pennsylvania lawmakers told supporters that the state should have more responsibility developing and implementing environmental policy. The comments came at a press conference last week where during which the politicians outlined their policy views.

Members of the group said that because the health of the environment is so important, the state and local governments should play the most prominent role in ensuring its protection. The senior member of the group noted, "It's about empowering the people. The federal government is too big and too slow to effectively handle an issue like this. State and local government are closer to the people and can do a better job deciding what's right for the environment."

The press conference was the first of many that are scheduled to take place in the next few months.

To what extent do you agree or disagree with the Pennsylvania lawmakers' thoughts on environmental policy?

Immigration Stimulus

Arizona lawmakers voiced their support for the "Support Our Law Enforcement and Safe Neighborhoods Act," considered to be one of the nation's broadest and strictest anti-illegal immigration measures. The act requires all aliens over the age of 14 to who remain in the United States for longer than 30 days to register with the US Government, and to have registration documents in their possession at all times. Additionally, the act requires that state law enforcement officers attempt to determine an individual's immigration status during a lawful stop, detention or arrest, or during a lawful contact not specific to any activity when there is reasonable suspicion that the individual is an illegal immigrant.

A junior congressman said that the act is popular among his constituents, and that it is "essential to protecting the jobs and the rights of American citizens."

The Arizona state legislature is set to discuss immigration policy in the next few months.

To what extent do you agree or disagree with the Arizona lawmakers' thoughts on immigration policy?

Social Desirability Battery (True/False)

1. It is sometimes hard for me to go on with my work if I am not encouraged.
2. I sometimes feel resentful when I don't get my way.
3. On a few occasions, I have given up doing something because I thought too little of my ability.
4. There have been times when I felt like rebelling against people in authority even though I knew they were right.
5. No matter who I'm talking to, I'm always a good listener.
6. There have been occasions when I took advantage of someone.
7. I'm always willing to admit it when I make a mistake.
8. I sometimes try to get even rather than forgive and forget.
9. I am always courteous, even to people who are disagreeable.
10. I have never been irked when people expressed ideas very different from my own.
11. There have times when I was quite jealous of the good fortune of others.
12. I am sometimes irritated by people who ask favors of me.
13. I have never deliberately said something that hurt someone's feelings.

General Emotion Regulation (1-7 Scale)

1. I tend to control my emotions most of the time.
2. Overall, I have a great deal of control over my emotions.

Emotion Reappraisal (1-7 Scale)

1. When I want to feel more positive emotion (such as joy or amusement), I change what I'm thinking about.
2. When I want to feel less negative emotion (such as sadness or anger), I change what I'm thinking about.
3. When I'm faced with a stressful situation, I make myself think about it in a way that helps me stay calm.
4. When I want to feel more positive emotion, I change the way I'm thinking about the situation.

5. I control my emotions by changing the way I think about the situation I'm in.
6. When I want to control my emotions, I'm not likely to change the way I think about the situation (reverse coded).
7. When I want to feel less negative emotion, I change the way I'm thinking about the situation.

Emotion Suppression (1-7 Scale)

1. I keep my emotions to myself.
2. When I am feeling positive emotions, I am careful not to express them.
3. I control my emotions by not expressing them.
4. I almost never try to inhibit my emotional expressions (reverse coded).
5. When I am feeling negative emotions, I make sure not to express them.